

MedKayfab: Microbiology & Infectiology Edition



BLÖDSINN!



1	A				
2	B				
3	C				
4	D				
5	E				

Stay
fluffy!



Gram the Stainer | The Gram Stain

STAT: 20-80nm (Thick Purple Wall)
vs 2-7nm (Thin Pink Wall)



POWER: The Great Divide: Instantly sorts all bacterial enemies into two factions, determining which empirical beta-lactam weapons will actually work.
LORE: Mycoplasmas, Mycobacteria, and Treponema laugh at your dyes. You cannot stain what has no normal wall.

KayfaBINGO!

Staph the Golden | Staphylococcus aureus

STAT: 30% (Asymptomatic
Nasal Carrier Rate)



POWER: Panton-Valentine Punch: Destroys incoming neutrophils to deal massive necrotising tissue damage.
LORE: S. aureus bacteraemia demands an echocardiogram. Every single time. No exceptions.

KayfaBINGO!

Strep the Streaker | Streptococcus Species

STAT: 2-4 Weeks
(Time to Rheumatic Fever onset)



POWER: Lancefield Shapeshift: Alters cell wall carbs to switch between causing strep throat, neonatal sepsis, and flesh-eating fasciitis.
LORE: Rheumatic fever only follows pharyngitis, NEVER skin infections.

KayNSINGO!

Neisseria the Socialite | N. meningitidis & N. gonorrhoeae

STAT: 100%
(Requires intimate/mucosal
contact to survive)



POWER: TLR4 Overdrive: Detonates LPS to cause Waterhouse-Friderichsen bilateral adrenal hemorrhage in hours.
LORE: Non-Elanching petechial rash + fever = IV Ceftriaxone IMMEDIATELY. Do not wait for the lumbar puncture.

Pseudo the Opportunist | *Pseudomonas aeruginosa*

STAT: 0 (Ertapenem's effectiveness against it)



POWER: Quorum Chat: Coordinates an irradicable mucoid biofilm shield when numbers reach critical mass.
LORE: It intrinsically resists most drugs; target with Pip-Tazo, Ceftazidime, or Meropenem. Never Ertapenem.

KayfaBINGO!

Clostridium the Anaerobe | *Clostridium* Species

STAT: 4 (Distinct apocalyptic toxins)



POWER: SNARE Snp: Tetanospasmin blocks inhibitory signals (spastic paralysis); Botulinum blocks acetylcholine (flaccid paralysis).
LORE: Heat kills the bacteria, but spores survive boiling. You must use the autoclave.

KayfaBINGO!

LPS the Endotoxin | Lipopolysaccharide

STAT: 1
 (The most dangerous molecule per microgram in medicine)



POWER: Jarisch-Herxheimer Surge: Detonates upon antibiotic death of spirochaetes, causing a massive 2-8 hour fever spike.
LORE: LPS causes DIC and distributive shock. Do NOT stop antibiotics during a Herxheimer reaction.



Biofilm Billy | Bacterial Biofilms

STAT: 1000x
 (Increased antibiotic tolerance vs planktonic cells)



POWER: Metabolic Dormancy: Slows growth to near-zero, rendering drugs that target active cell division completely useless.
LORE: You cannot cure a prosthetic joint or valve biofilm with drugs alone. The hardware must be removed.



Pneumo the Encapsulated | *Streptococcus pneumoniae* STAT: >90
(Different anti-phagocytic capsule serotypes)

POWER: Splenic Bypass: Causes overwhelming fulminant sepsis within hours if the host lacks marginal zone B cells (asplenia).
LORE: If the patient has no spleen, Pneumo is a terminal threat. Vaccinate and give prophylactic penicillin.

Listeria the Lurker | *Listeria monocytogenes* STAT: 4°C
(Optimal growth temperature)

POWER: Actin Rocket: Spreads from cell to cell without ever touching the extracellular space or facing an antibody.
LORE: Cephalosporins are useless here. Always add Amoxicillin for meningitis in patients >60 or pregnant.

TB the Persister | *Mycobacterium tuberculosis* STAT: 40,000 Years
(Co-evolution with humans)

POWER: Phagosome Hack: Uses ESX-1 to prevent lysosome fusion, surviving happily inside the very cells meant to digest it.
LORE: Never treat active TB with one drug. RIPE (4 drugs) prevents the rapid emergence of resistance.

MRSA the Resistance Fighter | *Methicillin-Resistant S. aureus* STAT: AUC/MIC 400-600
(Vancomycin target)

POWER: PBP2a Morph: Alters its penicillin-binding proteins so standard beta-lactams have zero affinity.
LORE: Daptomycin kills MRSA in the blood, but pulmonary surfactant inactivates it. Never use it for MRSA pneumonia.

E. coli the Versatile | *Escherichia coli*

STAT: 80%
(Cause of community UTIs)



POWER: Shiga Strike: Inhibits the 60S ribosome, causing endothelial death and Hemolytic Uremic Syndrome (HUS).
LORE: Never give antibiotics for EHEC (O157:H7). It triggers massive toxin release and worsens HUS.

Klebsiella the Fortress | *Klebsiella pneumoniae*

STAT: >5mm
(Positive 'String Test' stretch length)



POWER: Endogenous Seeding: Hypervirulent strains metastasize from liver abscesses directly into the eye.
LORE: KPC production destroys nearly all beta-lactams, requiring Ceftazidime-Avibactam combos to survive.

Salmonella the Traveller | *Salmonella* species

STAT: 7-21 Days
(Typhoid incubation period)



POWER: Macrophage Trojan: Hides in the reticuloendothelial system, causing a stepwise fever and relative bradycardia.
LORE: Treat Typhoid with antibiotics. Do NOT treat healthy non-typhoidal gastroenteritis with antibiotics (prolongs carriage).

Helicobacter the Ulcer Maker | *Helicobacter pylori*

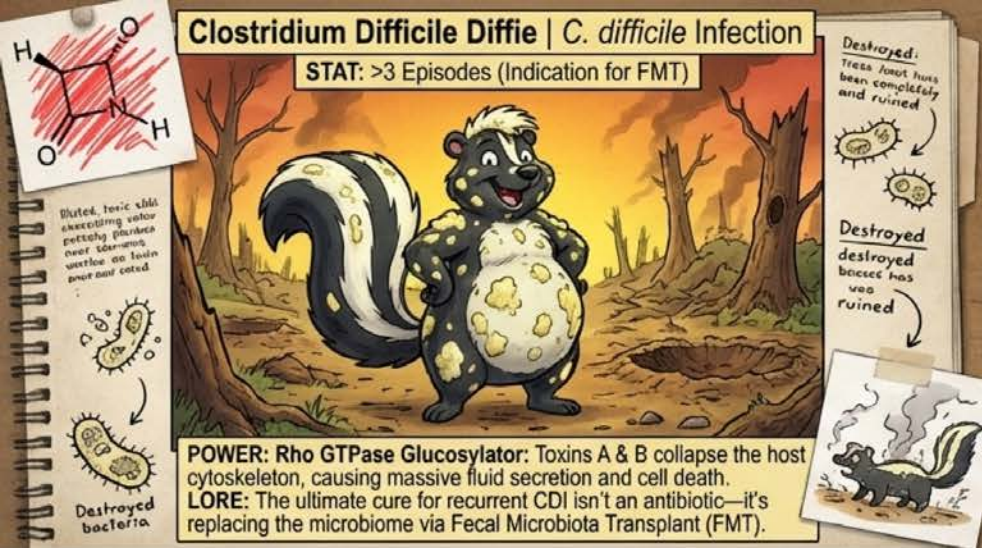
STAT: 50%
(Global human carrier rate)



POWER: Oncogenic Hack: Over decades, CagA disrupts epithelial junctions, directly driving gastric adenocarcinoma and MALT lymphoma.
LORE: Eradicating *H. pylori* causes complete remission in 75% of low-grade gastric MALT lymphomas.

Clostridium Difficile Diffie | C. difficile Infection

STAT: >3 Episodes (Indication for FMT)



Disturbed, toxic stool
unrelenting yellow
pale yellow diarrhea
over 30-40 years
worsens as toxin
poor our cated

Destroyed bacteria

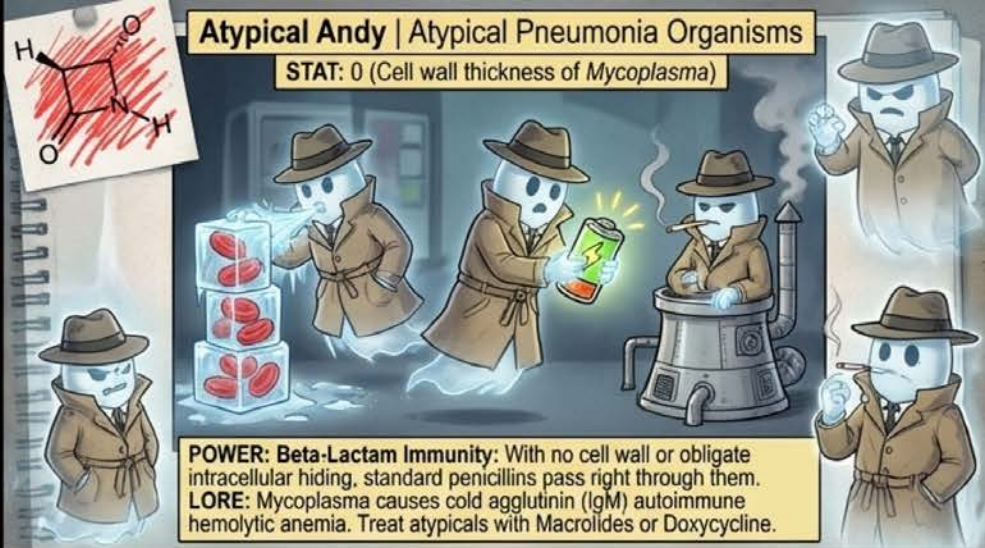
Destroyed:
These host cells
have been completely
and ruined

Destroyed
destroyed
bacteria has
been ruined

POWER: Rho GTPase Glucosylator: Toxins A & B collapse the host cytoskeleton, causing massive fluid secretion and cell death.
LORE: The ultimate cure for recurrent CDI isn't an antibiotic—it's replacing the microbiome via Fecal Microbiota Transplant (FMT).

Atypical Andy | Atypical Pneumonia Organisms

STAT: 0 (Cell wall thickness of Mycoplasma)



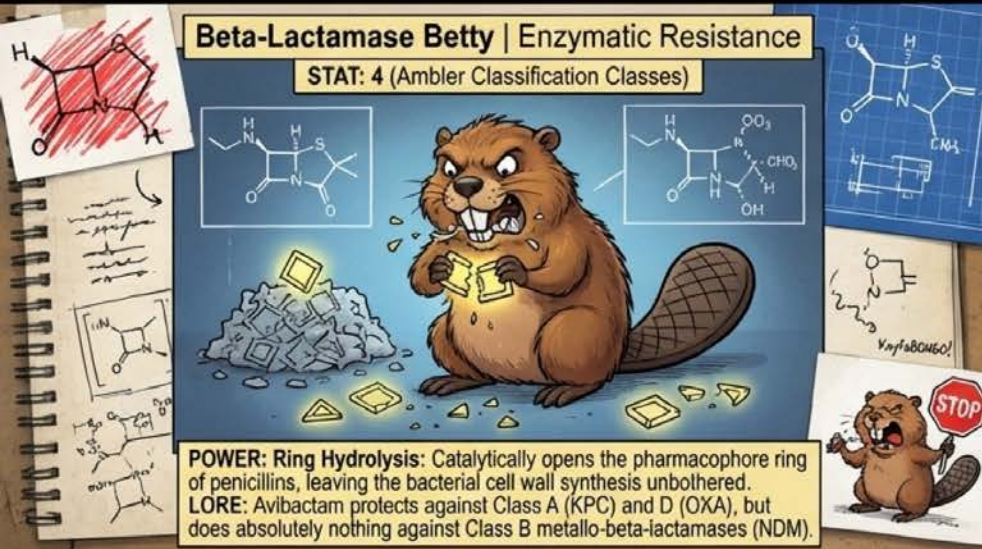
Destroyed:
These host cells
have been completely
and ruined

Destroyed
destroyed
bacteria has
been ruined

POWER: Beta-Lactam Immunity: With no cell wall or obligate intracellular hiding, standard penicillins pass right through them.
LORE: Mycoplasma causes cold agglutinin (IgM) autoimmune hemolytic anemia. Treat atypicals with Macrolides or Doxycycline.

Beta-Lactamase Betty | Enzymatic Resistance

STAT: 4 (Ambler Classification Classes)



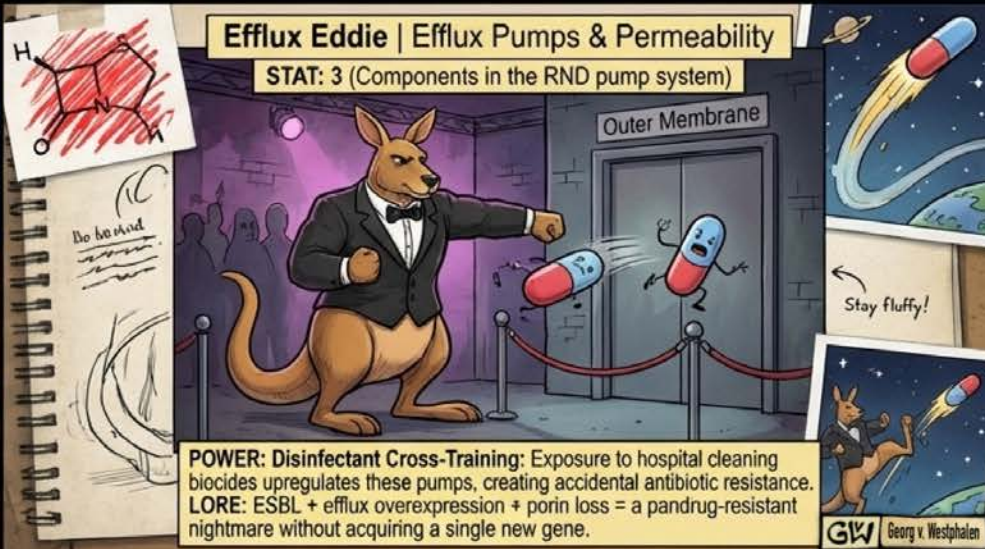
Destroyed:
These host cells
have been completely
and ruined

Destroyed
destroyed
bacteria has
been ruined

POWER: Ring Hydrolysis: Catalytically opens the pharmacophore ring of penicillins, leaving the bacterial cell wall synthesis unbothered.
LORE: Avibactam protects against Class A (KPC) and D (OXA), but does absolutely nothing against Class B metallo-beta-lactamases (NDM).

Efflux Eddie | Efflux Pumps & Permeability

STAT: 3 (Components in the RND pump system)



Destroyed:
These host cells
have been completely
and ruined

Destroyed
destroyed
bacteria has
been ruined

POWER: Disinfectant Cross-Training: Exposure to hospital cleaning biocides upregulates these pumps, creating accidental antibiotic resistance.
LORE: ESB + efflux overexpression + porin loss = a pandrug-resistant nightmare without acquiring a single new gene.

HGT the Gene Trader | Horizontal Gene Transfer

STAT: 1
(One Health Framework)

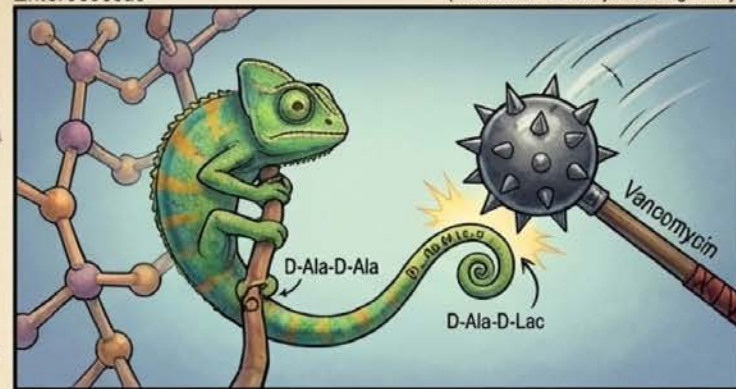


POWER: Conjugation Pilus: Transfers multidrug resistance across entirely different bacterial species in a single physical handshake.
LORE: A patient who has never taken antibiotics can get an ESBL *E. coli* because veterinary and agricultural use breeds resistance that spreads laterally.

KofoBIZARRO!

VRE the Glycopeptide Evader | Vancomycin-Resistant Enterococcus

STAT: 1000x
(Reduction in Vancomycin binding affinity)



POWER: Terminal Substitution: Rewrites its own cell wall precursor, retaining structural integrity while entirely evading glycopeptide blockade.
LORE: VRE endocarditis is highly lethal. Switch to Daptomycin (high dose) or Linezolid immediately.

HUMBUG!

Steward Stewart | Antibiotic Stewardship

STAT: ~15
(New systemics approved since 2000)



POWER: De-escalation Review: Strips away empiric broad-spectrum armor at 48-72h, switching to sniper-rifle targeted therapy based on culture data.
LORE: Shorter is better. A 5-day course for community-acquired pneumonia is just as effective as 10 days, and prevents collateral microbiome damage.

KofoBIZARRO!

Carbapenem Carl | Carbapenems

STAT: 30-50%
(30-day mortality of CRE infections)



POWER: Broad-Spectrum Nuke: Hits Gram-positives, Gram-negatives, Pseudomonas, and anaerobes simultaneously.
LORE: Ertapenem is the only member of the gang that cannot kill Pseudomonas. Don't send it to do Meropenem's job.

KofoBIZARRO!

HUMBUG!

ESKAPE the Gang | The Hospital Resistance Ecosystem

STAT: 6
(The ultimate hospital bad guys)

KoyfaBOGGLE!



POWER: Inducible AmpC: Enterobacter pretends to be susceptible to 3rd-gen cephalosporins, then mutates mid-treatment to destroy them.
LORE: Contact precautions (gloves/aprons) and hand hygiene are the only way to stop this gang from running the ICU.

Sepsis Sam | The Sepsis Cascade

STAT: 22
(SOFA score jump defining Sepsis)

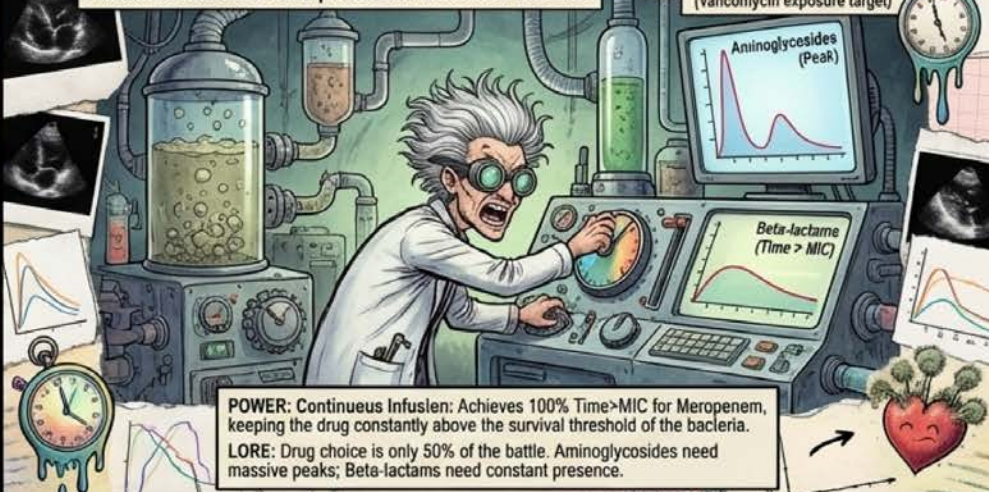


POWER: Endothelial Leak: Inflammatory cytokines degrade the glycocalyx, turning closed blood vessels into a porous, deadly sponge.
LORE: Sepsis kills through the dysregulated host response, not the pathogen directly. Lactate >2 indicates tissue hypoxia.

BLÖDSINN!

Pharmacokinetics Petra | PK/PD in Infectious Disease

STAT: AUC 400-600
(Vancomycin exposure target)



POWER: Continuous Infusion: Achieves 100% Time>MIC for Meropenem, keeping the drug constantly above the survival threshold of the bacteria.
LORE: Drug choice is only 50% of the battle. Aminoglycosides need massive peaks; Beta-lactams need constant presence.

Endocarditis Eddie | Infective Endocarditis

STAT: >10mm
(Vegetation size risking embolization)



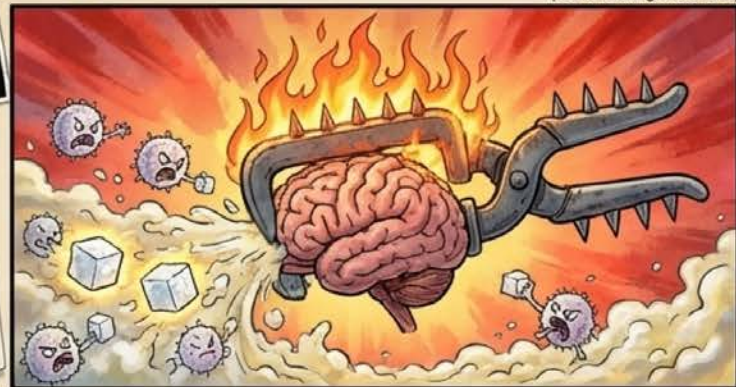
POWER: Sterile Thrombus Base: Requires pre-existing turbulent flow or valve damage to create a sticky landing pad for passing bacteria.
LORE: Prosthetic valve endocarditis requires Rifampicin for biofilm penetration, but NEVER as monotherapy.

BLÖGW

Georg v. Westphalen

Meningitis Mike | Bacterial Meningitis

STAT: <0.4
(CSF:Plasma glucose ratio)



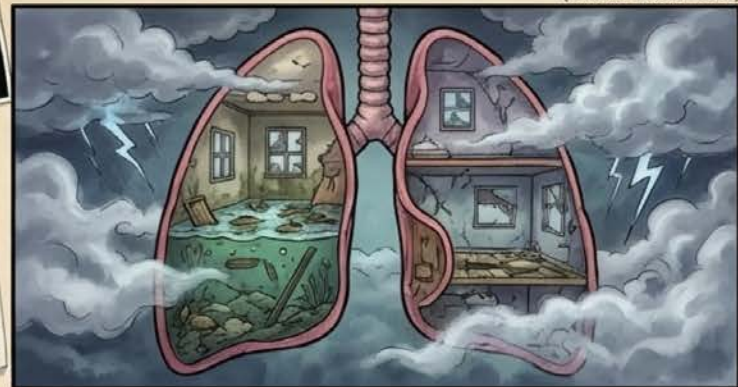
POWER: Endocochlear Inflammation: Causes severe, permanent hearing loss if not blunted by steroids.

LORE: Dexamethasone reduces mortality and hearing loss, but ONLY if given 15-30 minutes before or with the first antibiotic.

KayfaBIZARRO!

Pneumonia Petra | Community-Acquired Pneumonia

STAT: CURB-65 ≥ 3
(Consider ICU admission)



POWER: Severity Stratification: Matches the clinical care setting perfectly to the objective risk of death.

LORE: Amoxicillin + Clarithromycin empirically covers typicals and atypicals. Add Levofloxacin if Legionella (hyponatremia + confusion) is suspected.

HUMBUG!

What the FLUFF?!

UTI Ursula | Urinary Tract Infections

STAT: $\geq 10^5$ CFU/mL
(Definition of significant bacteriuria)



POWER: Urease Stonecrafter: Protsus uses urease to create an alkaline environment, forging massive staghorn struvite calculi.

LORE: DO NOT TREAT asymptomatic bacteriuria in non-pregnant, non-surgical patients. It only selects for resistance.

Kayfa-BINGO!

Septic Arthritis Sally | Bone & Joint Infections

STAT: 48-72h
(Time to irreversible cartilage destruction)



POWER: Synovial Invasion: Bypasses standard defenses because the highly vascular synovial membrane lacks a basement membrane.

LORE: Antibiotics are not enough. A septic joint is a surgical emergency requiring immediate drainage to stop protease destruction.

What the FL GW Georg v. Westphalen

Meningococcal Monica | Meningococcal Disease

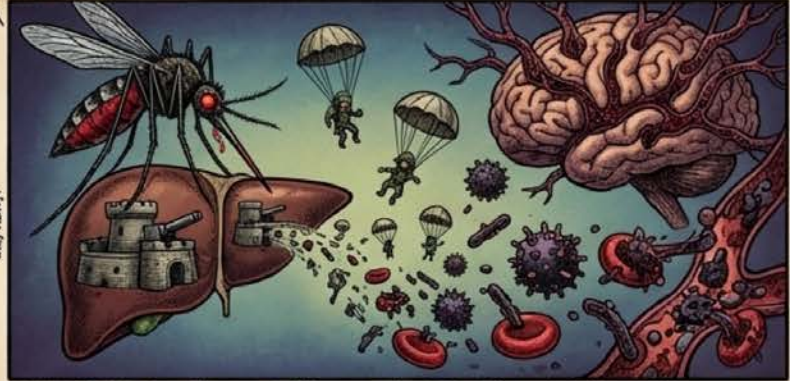
STAT: 12-24 Hours
(Time from symptoms to death)



POWER: DIC Detonation: Microvascular thrombosis leads directly to tissue ischemia and adrenal hemorrhage.
LORE: Speed is everything. Give IM Benzylpenicillin or IV Ceftriaxone immediately; do not wait for tests. The rash means DIC is already happening.

Tropical Fever Theo | Malaria

STAT: 48h / 72h
(Erythrocytic schizogony fever cycles)



POWER: PfEMP1 Cytoadherence: *P. falciparum* makes RBCs stick to endothelium, causing coma (cerebral malaria) and avoiding splenic clearance.
LORE: You must give Primaquine to cure *P. vivax/ovale* hypnozoites in the liver, but only after checking for G6PD deficiency to avoid massive hemolysis.

HIV the Retrovirus | Immunopathogenesis & ART

STAT: <200 cells/ μ L
(AIDS-defining CD4 threshold)



POWER: Error-Prone Replication: Mutates so fast that single-drug therapy guarantees instant resistance.
LORE: Undetectable = Untransmittable (U=U). However, prophylaxis (e.g., co-trimoxazole for PCP) must continue until the CD4 count recovers above 200.

Fungal Fran | Invasive Fungal Infections

STAT: >25 cmH₂O
(Critical opening pressure in Cryptococcus)



POWER: Host-Defect Matcher: Only attacks specific immune gaps (e.g., *Aspergillus* targets neutropenia; *Cryptococcus* targets CD4 <100).
LORE: Candidaemia requires mandatory fundoscopy and echocardiography, and the central line must be removed immediately.

Immunocompromised Ian | The Defective Host

STAT: ANC <500
(Definition of severe neutropenia)

POWER: Febrile Emergency: A temperature in a neutropenic patient bypasses all waiting rooms directly to IV Pip-Tazo/Meropenem.
LORE: The specific immune defect perfectly predicts the organism. Asplenia = Encapsulated bacteria. T-Cell defect = Intracellular pathogens.

KeyInBUNGO!

Viral Hepatitis Vector | Hepatitis B & C

STAT: >95%
(SVR/Cure rats for Hep C with DAAs)

POWER: Perinatal Persistence: 90% of perinatally infected Hep B infants become chronic carriers, compared to 5% of adults.
LORE: Tenofovir suppresses Hep B but rarely cures it (due to cccDNA). DAAs completely cure Hep C in 8-12 weeks.

KeyInBUNGO!

Hand Hygiene Hannah | Infection Prevention & Control

STAT: 5
(WHO Moments for Hand Hygiene)

POWER: Chain Breaker: The single most statistically effective intervention in the history of hospital medicine.
LORE: Alcohol rub kills 99.99% of things, but does absolutely nothing to *C. difficile* spores or Noroviruses. Soap and mechanical friction are required.

KeyInBUNGO!

Zoonosis Zara | Animal-to-Human Transmission

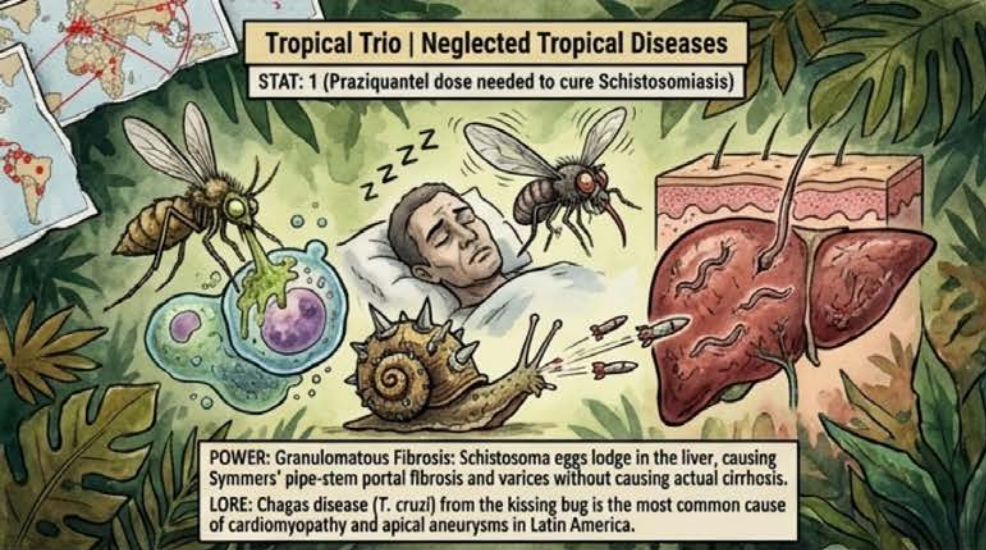
STAT: >Scm
(Erythema migrans diagnostic size for Lyme)

POWER: Vector Jump: Uses an intermediate animal host to bypass human-to-human transmission limits.
LORE: Pasteurella from a cat/dog bite causes rapid cellulitis. Treat empirically with Amoxicillin-Clavulanate, not standard Flucloxacillin.

GW Georg v. Westphalen

Tropical Trio | Neglected Tropical Diseases

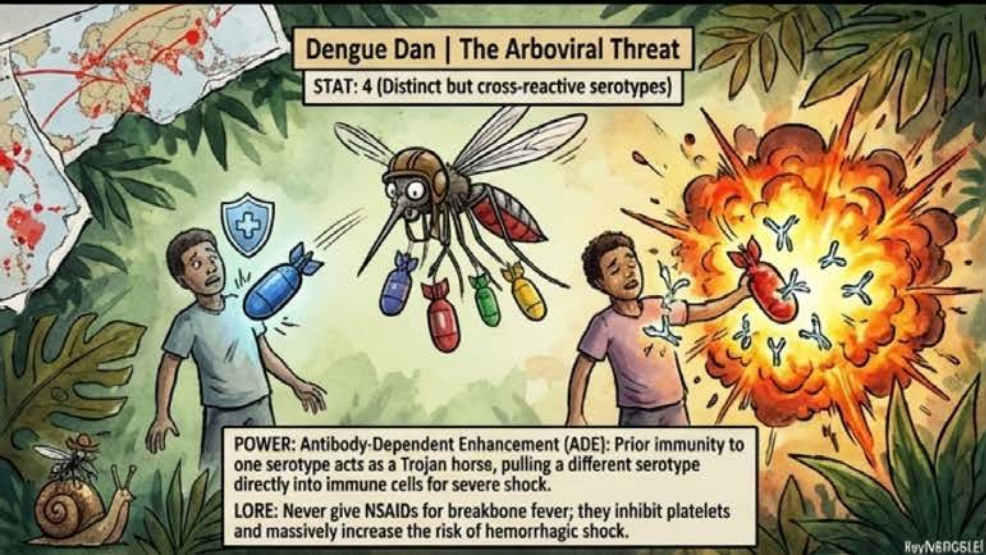
STAT: 1 (Praziquantel dose needed to cure Schistosomiasis)



POWER: Granulomatous Fibrosis: Schistosoma eggs lodge in the liver, causing Symmers' pipe-stem portal fibrosis and varices without causing actual cirrhosis.
LORE: Chagas disease (*T. cruzi*) from the kissing bug is the most common cause of cardiomyopathy and apical aneurysms in Latin America.

Dengue Dan | The Arboviral Threat

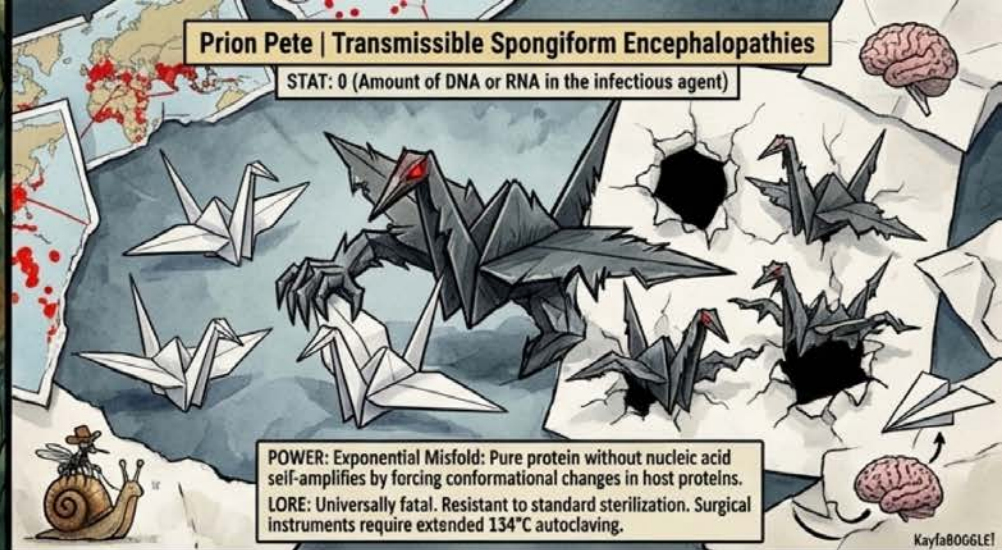
STAT: 4 (Distinct but cross-reactive serotypes)



POWER: Antibody-Dependent Enhancement (ADE): Prior immunity to one serotype acts as a Trojan horse, pulling a different serotype directly into immune cells for severe shock.
LORE: Never give NSAIDs for breakbone fever; they inhibit platelets and massively increase the risk of hemorrhagic shock.

Prion Pete | Transmissible Spongiform Encephalopathies

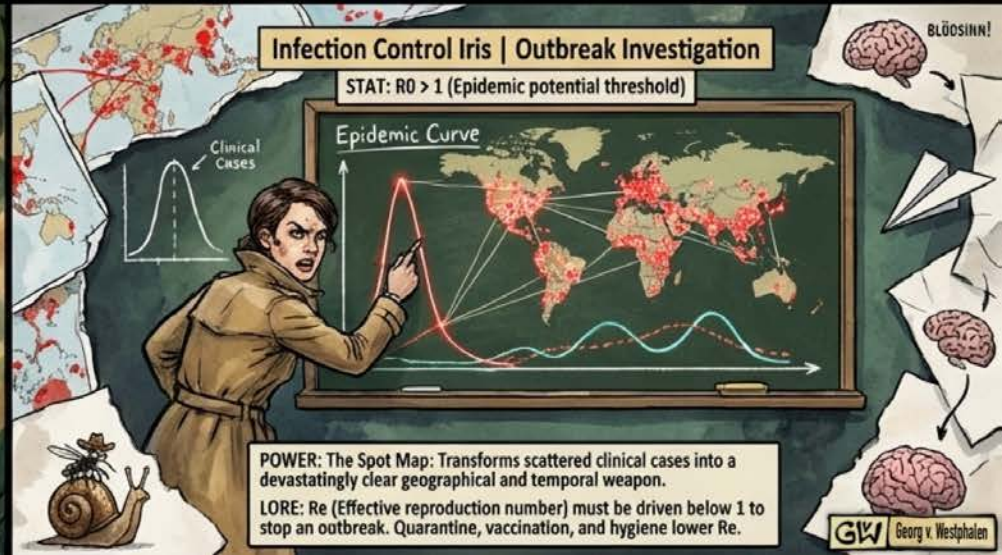
STAT: 0 (Amount of DNA or RNA in the infectious agent)



POWER: Exponential Misfold: Pure protein without nucleic acid self-amplifies by forcing conformational changes in host proteins.
LORE: Universally fatal. Resistant to standard sterilization. Surgical instruments require extended 134°C autoclaving.

Infection Control Iris | Outbreak Investigation

STAT: $R_0 > 1$ (Epidemic potential threshold)

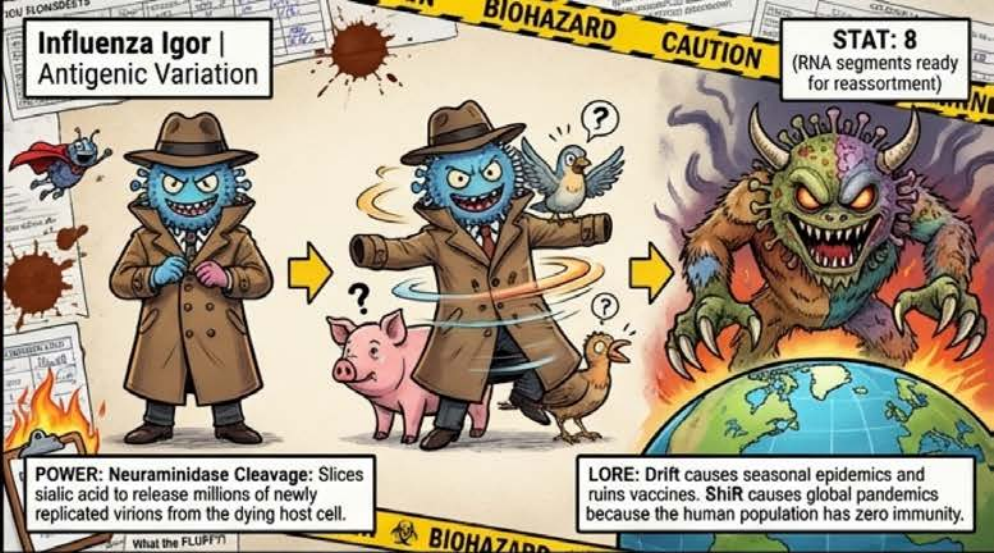


POWER: The Spot Map: Transforms scattered clinical cases into a devastatingly clear geographical and temporal weapon.
LORE: R_e (Effective reproduction number) must be driven below 1 to stop an outbreak. Quarantine, vaccination, and hygiene lower R_e .

Candida Auris the Superbug | Emerging MDR Fungal Pathogen



Influenza Igor | Antigenic Variation



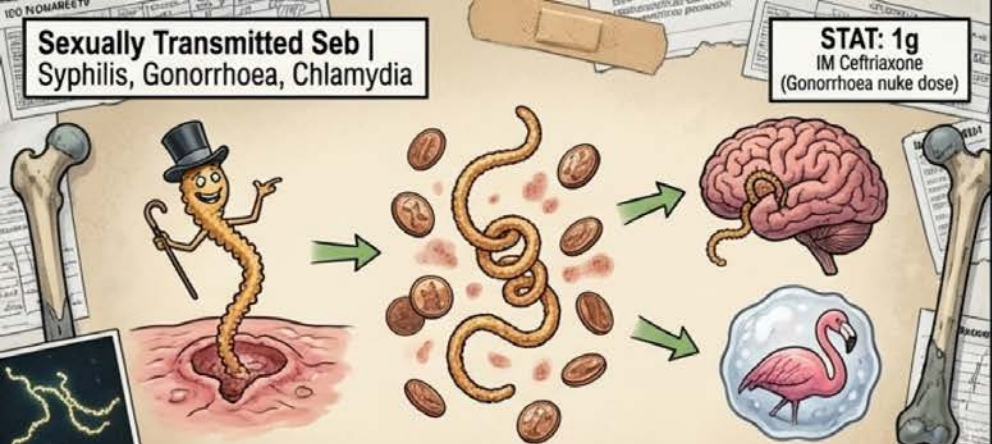
Sepsis Six Sienna | The Hour-1 Bundle



Tropical Neglected Nelly | Post-Travel Assessment



Sexually Transmitted Seb | Syphilis, Gonorrhoea, Chlamydia



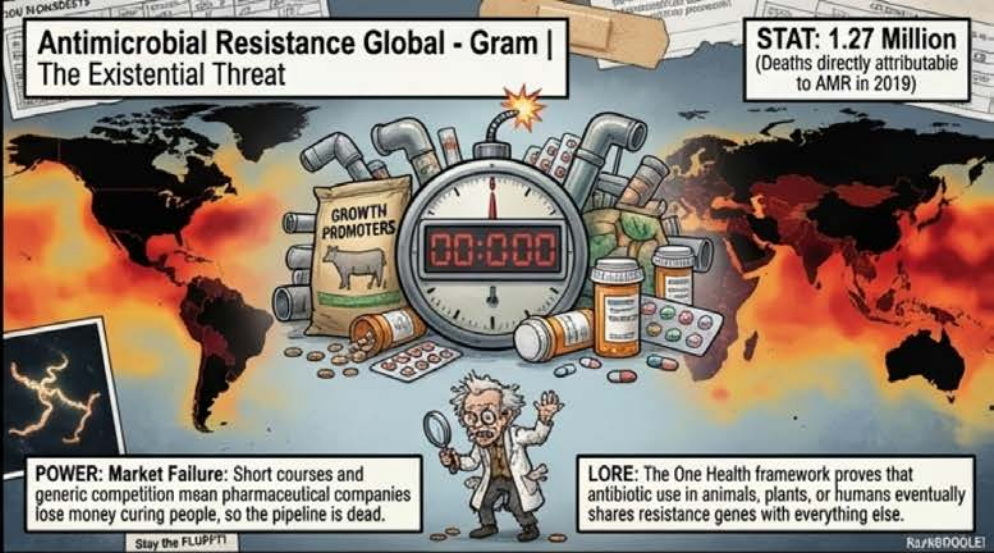
STAT: 1g
IM Ceftriaxone
(Gonorrhoea nuke dose)

POWER: Argyll Robertson Pupil: Tertiary neurosyphilis damages the dorsal midbrain—the pupils accommodate to distance, but refuse to react to light.

LORE: Treponemal tests stay positive for life. Non-treponemal tests go down with treatment. Treat all stages with Penicillin.

Antimicrobial Resistance Global - Gram | The Existential Threat

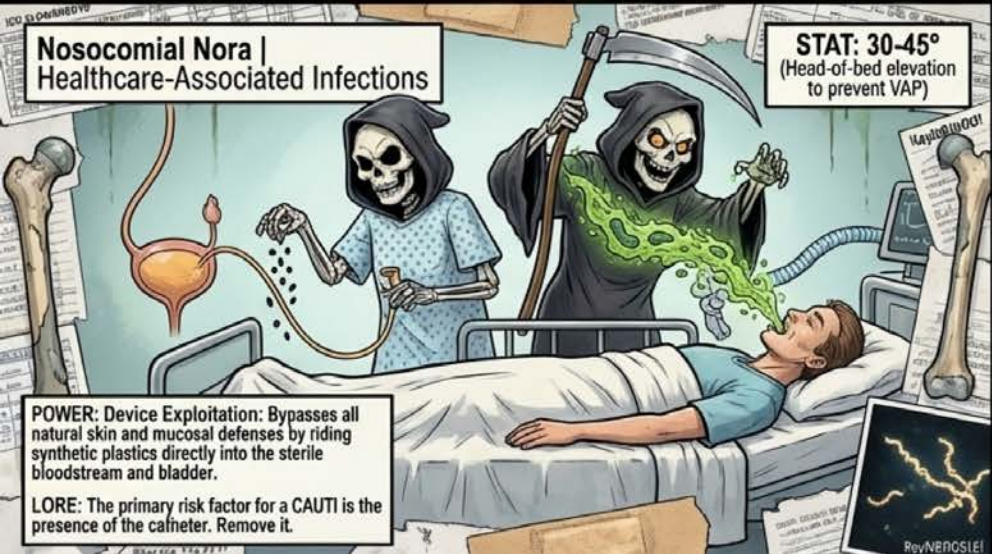
STAT: 1.27 Million
(Deaths directly attributable to AMR in 2019)



POWER: Market Failure: Short courses and generic competition mean pharmaceutical companies lose money curing people, so the pipeline is dead.

LORE: The One Health framework proves that antibiotic use in animals, plants, or humans eventually shares resistance genes with everything else.

Nosocomial Nora | Healthcare-Associated Infections



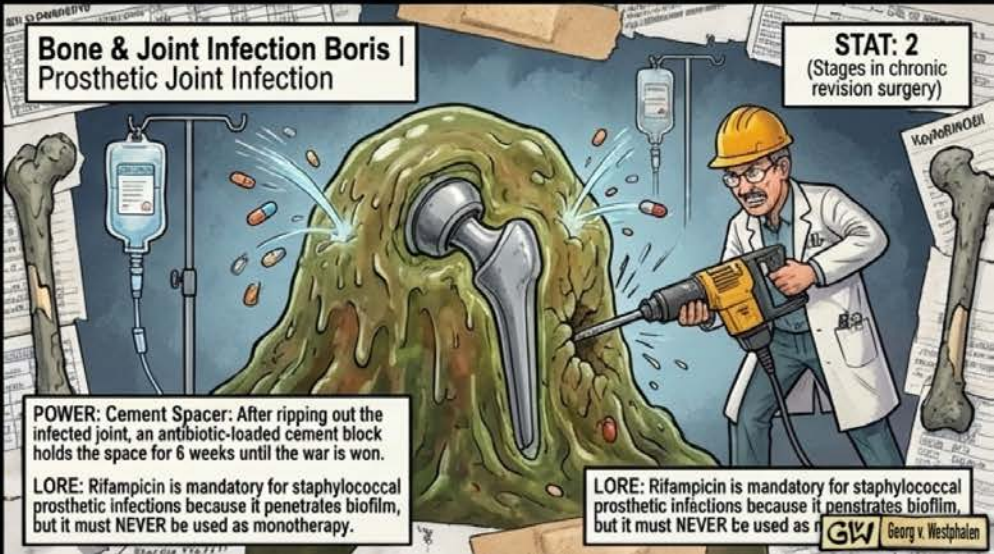
STAT: 30-45°
(Head-of-bed elevation to prevent VAP)

POWER: Device Exploitation: Bypasses all natural skin and mucosal defenses by riding synthetic plastics directly into the sterile bloodstream and bladder.

LORE: The primary risk factor for a CAUTI is the presence of the catheter. Remove it.

Bone & Joint Infection Boris | Prosthetic Joint Infection

STAT: 2
(Stages in chronic revision surgery)



POWER: Cement Spacer: After ripping out the infected joint, an antibiotic-loaded cement block holds the space for 6 weeks until the war is won.

LORE: Rifampicin is mandatory for staphylococcal prosthetic infections because it penetrates biofilm, but it must NEVER be used as monotherapy.

LORE: Rifampicin is mandatory for staphylococcal prosthetic infections because it penetrates biofilm, but it must NEVER be used as monotherapy.

GW Georg v. Westphalen

Infection in Pregnancy Irene TORCH Syndromes

STAT: 15-30%
(Maternal GBS carriage rate)

Antifungal Andy | Antifungal Pharmacology

STAT: CYP3A4
(The metabolic enzyme destroyed by Azoles)



POWER: First Trimester Teratogeny: The earlier the virus crosses the placenta during organogenesis, the more devastating the anatomical destruction.

LORE: Rubella is a live vaccine. Never give it during pregnancy. CMV is the most common cause of congenital sensorineural hearing loss.



POWER: Drug-Interaction Nuke: Fluconazole and Voriconazole inhibit CYP enzymes so hard that co-administered transplant drugs (Tacrolimus) spike to toxic levels.

LORE: Echinocandins are empirical first-line for critically ill candidaemia because they ignore CYP interactions and kill azole-resistant strains.

The ID Attending | Master Synthesis Card

STAT: 100%
(Integration required)

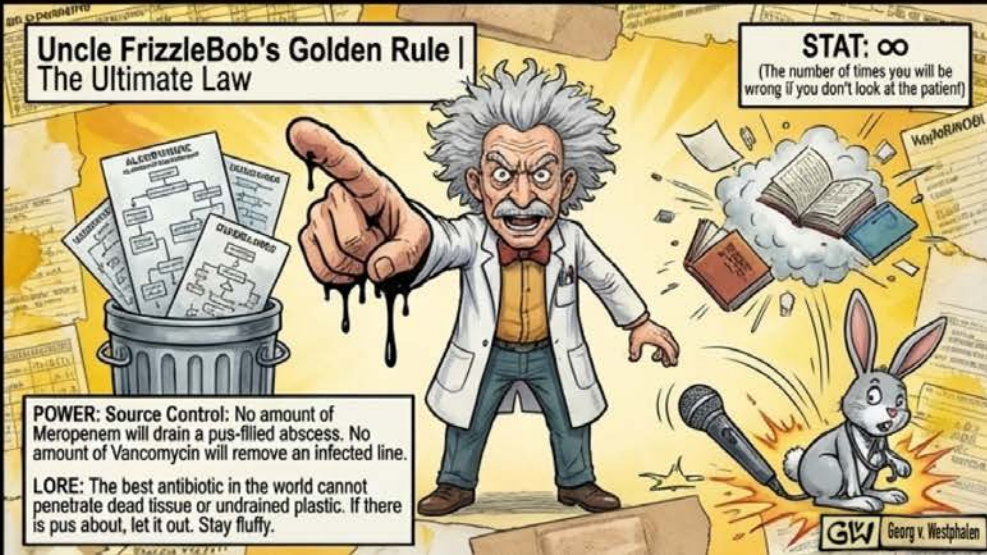
Uncle FrizzleBob's Golden Rule | The Ultimate Law

STAT: ∞
(The number of times you will be wrong if you don't look at the patient)



POWER: Clinical Reasoning: Identifies the immune defect (Myeloma), correlates the organism (*S. aureus*), applies the criteria (Echo + Bloods = Endocarditis), and defines the surgery indication.

LORE: An isolated fact is useless. You must link the anatomical site, the immune defect, the resistance mechanism, and the pharmacokinetics to save the life.



POWER: Source Control: No amount of Meropenem will drain a pus-filled abscess. No amount of Vancomycin will remove an infected line.

LORE: The best antibiotic in the world cannot penetrate dead tissue or undrained plastic. If there is pus about, let it out. Stay fluffy.